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Delay Prediction Through Multi-Channel Traffic and Weather Scene Image: A Deep Learning-Based Method

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Abstract

Accurate prediction of airport delays under convective weather conditions is essential for effective traffic coordination and improving overall airport efficiency. Traditional methods mainly rely on numerical weather and traffic indicators, but they often fail to capture the spatial distribution of traffic flows within the terminal area. To address this limitation, we propose a novel image-based representation named Multi-Channel Traffic and Weather Scene Image (MTWSI), which maps both meteorological and traffic information onto a two-dimensional airspace grid, thereby preserving spatial relationships. Based on the MTWSI, we develop a delay prediction model named ADLCNN. This model first uses a convolutional neural network to extract deep spatial features from the scene images and then classifies each sample into a delay level. Using real operational data from Guangzhou Baiyun Airport, this paper shows that ADLCNN achieves significantly higher prediction accuracy compared to traditional machine learning methods. The results confirm that MTWSI provides a more accurate representation of real traffic conditions under convective weather.

Keywords: airport delay prediction; image recognition; convolutional neural network (CNN); convective weather; class imbalance

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1. Introduction

Airport delay serves as a key indicator of an airport's overall operational performance, reflecting its efficiency and the effectiveness of flight scheduling. Various factors, such as air traffic control, weather conditions, aircraft maintenance, flight scheduling, airport facilities, and human factors, work together to cause airport delays. Serious delays can lead to flight congestion, a higher risk of conflicts or accidents, and major inconvenience for airlines, passengers, and airport operations. To reduce these negative effects, intelligent airport delay prediction systems have emerged as a key part of Intelligent Air Traffic Management (IATM). These systems aim to improve operational efficiency by using Artificial Intelligence (AI) for real-time monitoring and early warnings of flight delays.

The impact of weather on airport delays is particularly pronounced, especially when convective weather in terminal areas, such as high winds, hailstorms, or thunderstorms, disrupts aircraft operations and ground equipment. Therefore, an effective airport delay prediction system must take weather conditions into account. The primary objective of this paper is to characterize the influence of terminal area convective weather on flight operations by employing a multi-channel pictorial method that accurately depicts the

relative positions of convective weather and inbound/outbound flight flows, thereby improving existing delay prediction models in terms of both accuracy and practicality.

Compared to previous research, this study introduces a novel data representation method by rasterizing terminal area convective weather and flight data into Multi-Channel Traffic and Weather Scene Images. A Convolutional Neural Network (CNN) is then used to extract complex delay-related features from these images, allowing the development of a high-precision airport delay prediction model. The structure of the paper is as follows: Section 2 reviews related work on the weather delay index and deep prediction models. Section 3 explains the two main methods. Section 4 introduces the dataset and evaluates the proposed model through comparative experiments. Section 5 summarizes the study and outlines future research directions.

2. Literature Review

2.1. Related Work

2.1.1. Research on Weather Delay Index

Since the late 1990s, airport delay studies in the context of severe weather have been at the forefront of research, with pioneering work led by organizations such as MITRE, MIT Lincoln Laboratory, and NASA. Early studies focused on the Weather Impacted Index (WITI) developed by MITRE, along with related initiatives like MITRE's Flow Contingency Management (FCM), the application of Ensemble Weather Forecasts (EWF) to quantify weather uncertainty, and the FAA-led Air Traffic Management–Weather Integration study.

In terms of WITI, a great deal of research has been done by scholars at home and abroad. Callaham et al. [1] introduced WITI in 2001 to quantify how unfavorable weather affects the National Airspace System, particularly for large-scale or long-duration scenarios. Between 2007 and 2009, Sridhar and Chatterji [2,3] developed regression models that incorporated weather conditions, flight traffic, and airport size to more comprehensively analyze daily national airspace delays. Klein et al. [4,5] further refined the concept by proposing the enroute WITI (E-WITI), the terminal area WITI (T-WITI), queuing delays, and a predictive WITI FA index, and they introduced a scanline-based method (2008–2009) for assessing sector availability under convective weather forecasts [6,7]. In 2009, Cook et al. [8] quantified the weather's effect on air traffic via WITI. Mukherjee et al. [9] used WITI to classify weather days and compare weather-induced rerouting with National Advanced Planning Manual guidelines [10] developed a WITI-based method to evaluate sector capacity under severe weather.

2.1.2. Research on Deep Prediction Models

In recent years, Deep learning has rapidly transformed both weather prediction and traffic flow prediction. In 2020, Weyn et al. [11] introduced a data-driven global weather prediction framework using a deep CNN to fuse multichannel weather data, significantly boosting prediction accuracy. Building on this, Dong Kyun Kim et al. [12] combined ConvLSTM with 3D CNN for precipitation nowcasting, demonstrating deep models' strength in capturing spatiotemporal dynamics, while Jesús García Fernández et al. [13] proposed Broad UNet to reduce parameter counts without sacrificing performance in nowcasting. In 2022, Fowdur et al. [14] advanced a real-time collaborative machine learning framework that integrates distributed data sources to improve situational awareness for convective weather. More recently, Zhang J et al. [15] developed a CNN BiLSTM attention model that enhances severe convective weather forecasts, and Jardines et al. [16] created a CNN-based system for pre-tactical thunderstorm prediction to support flight scheduling. For very short-term predictions, Bouchard et al. [17] compared threshold methods, neural networks, and belief functions for lightning strike risk, and Gao J et al. (2023) [18] showed that a hy-

brid multichannel CNN-GRU model yields high precision in ultra-short-term wind power forecasting. Additionally, GraphCast [19] delivers global daily forecasts up to 10 days with higher accuracy and much lower cost than NWP; MetNet-2 provides 1 km, 12 h end-to-end forecasts, showcasing deep learning's extended lead-time edge [20].

In the traffic flow domain, traditional CNN/LSTM pipelines evolved toward graph architectures. Wang C et al. [21] introduce the Bo-GRA-CNN-BiLSTM model and employ Bayesian optimization for automatic hyperparameter tuning to enhance adaptability and accuracy under varying weather conditions. Cai Kaiquan et al. proposed a Geographical and Operational GCN, or multi-airport flight delay prediction, effectively capturing spatial–operational dependencies [22]. Qi et al. [23] proposed STFGCN, which fuses weather and traffic via spatial–temporal relationship modeling to achieve top performance across all weather conditions. Beyond Bayesian and Adam optimization methods, genetic algorithms and other evolutionary strategies are increasingly adopted for hyperparameter tuning in intelligent transportation systems. For instance, Akopov and Beklaryan [24] proposed a hybrid clustering-assisted genetic algorithm for the evolutionary synthesis of high-capacity multilayer road networks. Similarly, Priya and Francis [25] integrated a multi-objective genetic algorithm with RNNs to enhance traffic speed prediction accuracy in complex urban environments. Collectively, these advances illustrate deep learning's power to improve predictive accuracy and model design, while highlighting ongoing challenges in model complexity, data heterogeneity, and generalization.

2.2. Research Gap

Most existing research relies on statistical data to measure how convective weather affects traffic operations, mainly analyzing the overall effects in terminal areas. However, these studies have some important limitations:

(1) Do not take into account the spatial distribution of convective weather in the terminal area and its relative position to the flight trajectory.

(2) Current methods focus on single-airport or flight-level delay prediction, neglecting comprehensive evaluation of multi-airport collaborative delay levels under terminal-area operational scenarios.

(3) Conventional models rely on manually designed static features (e.g., visibility, wind speed), which inadequately capture spatial correlations between weather patterns and flight paths, resulting in suboptimal prediction accuracy.

We summarized the above studies in four aspects, including study area, research focus, key features, and unaddressed gaps, as shown in Table 1.

Therefore, this paper proposes a method for predicting airport delay level based on deep metric learning. The method focuses on airport delay levels as the main research object. First, the FCM algorithm is used to cluster delay-related features, extract classification rules for delay levels, and assign delay level labels. Then, actual data from the terminal area, including weather avoidance areas (WAF) and radar tracks (ADS-B), are converted into multi-channel image features that retain spatial location information. After that, a delay-level prediction model is built using a convolutional neural network. The aim of this method is to provide a comprehensive and objective way to describe and monitor the overall operational status of the terminal area. By accurately predicting the airport's delay conditions during convective weather, the method aims to support traffic management decisions in the pre-tactical phase and improve the operational efficiency of the terminal area.

Table 1. Summary of research on airport delay prediction.

Study Area	Research Focus	Key Features	Unaddressed Gaps
Weather Delay Index (WITI)	WITI, E-WITI, FCM (Callaham, Sridhar, Klein [1–5], Cook et al. [8])	Statistical indicators; regression models; sector-level delay estimates under convective weather	① Lacks spatially explicit weather trajectory modeling; ② Focus on national/airport-scale trends, limited flight-level granularity
Deep Learning for Weather Forecasting	CNN, ConvLSTM, Broad UNet, CNN-BiLSTM-Attention (Weyn et al. [11], Kim et al. [12], Zhang et al. [15])	Multichannel radar/satellite input; spatiotemporal dynamics; attention mechanisms	① Focus on pure weather prediction, not directly linked to flight trajectory or delay; ② Lack of operational integration in multi-airport scenarios
Deep Learning for Traffic/Delay Prediction	Bo-GRA-CNN-BiLSTM, Geographical GCN, STFGCN (Wang C, Cai K, Qi et al. [21–23])	Graph-based spatial–temporal modeling; hyperparameter tuning; weather–traffic fusion	① Most models use structured tabular data (visibility, wind, etc.), not unstructured image data; ② Spatial correlations between convective patterns and trajectory paths are weakly modeled
Integrated Traffic–Weather Learning Models	Hybrid CNN-GRU; GraphCast; MetNet-2 [18–20]	Ultra-short-term, high-resolution forecasts with multiscale features	① Limited exploration of real-scene imagery (e.g., radar/traffic images) in delay prediction tasks; ② Models seldom generalize across airport networks under dynamic weather scenes

3. Methodology

3.1. Multi-Channel Traffic and Weather Scene Image

Predicting airport delay levels under convective weather is challenging because traditional numerical features cannot fully capture the complex impact on terminal flight operations [24]. To address this issue, we apply deep learning by converting air traffic scene data into multi-channel images, from which useful features can be extracted. Figure 1 shows the overall framework for constructing the Multi-Channel Traffic and Weather Scene Image (MTWSI).

3.1.1. Data Description

This study uses Weather Advective Flow (WAF) data together with air traffic data, which are further categorized into dynamic flight data and static airspace information. WAF data represent convective cloud information captured by weather radar and are used to predict and monitor extreme weather events (e.g., torrential rains, thunderstorms, and tornadoes) [25]. Each WAF dataset consists of 1250 × 1250 data points, with each point indicating convective weather intensity over a 200 m × 200 m area. Static airspace data includes fixed aeronautical information, such as latitude and longitude, which define the structure of the airspace. Dynamic flight data, collected via satellite communication or ADS-B, provides real-time updates on aircraft position, altitude, speed, navigation, and flight status. In the proposed MTWSI-based deep learning method, the airspace is first rasterized using the static data. Then, the resulting grid is filled with dynamic flight data to form detailed traffic and weather scenarios. These images are then used for feature extraction to predict airport delay levels.

3.1.2. Construction of Images

(1) Rasterization of terminal areas

Convective weather in the terminal area is a major factor contributing to airport delays, especially when a high volume of flights is scheduled. In addition, the relative position

between the convective weather and flight paths plays a key role in determining the severity of the impact. For this reason, this paper proposes a new data representation method called multi-channel air traffic and weather scene images to portray the impact of convective weather within the terminal area on flight operations and to serve as an input to airport delay prediction models.

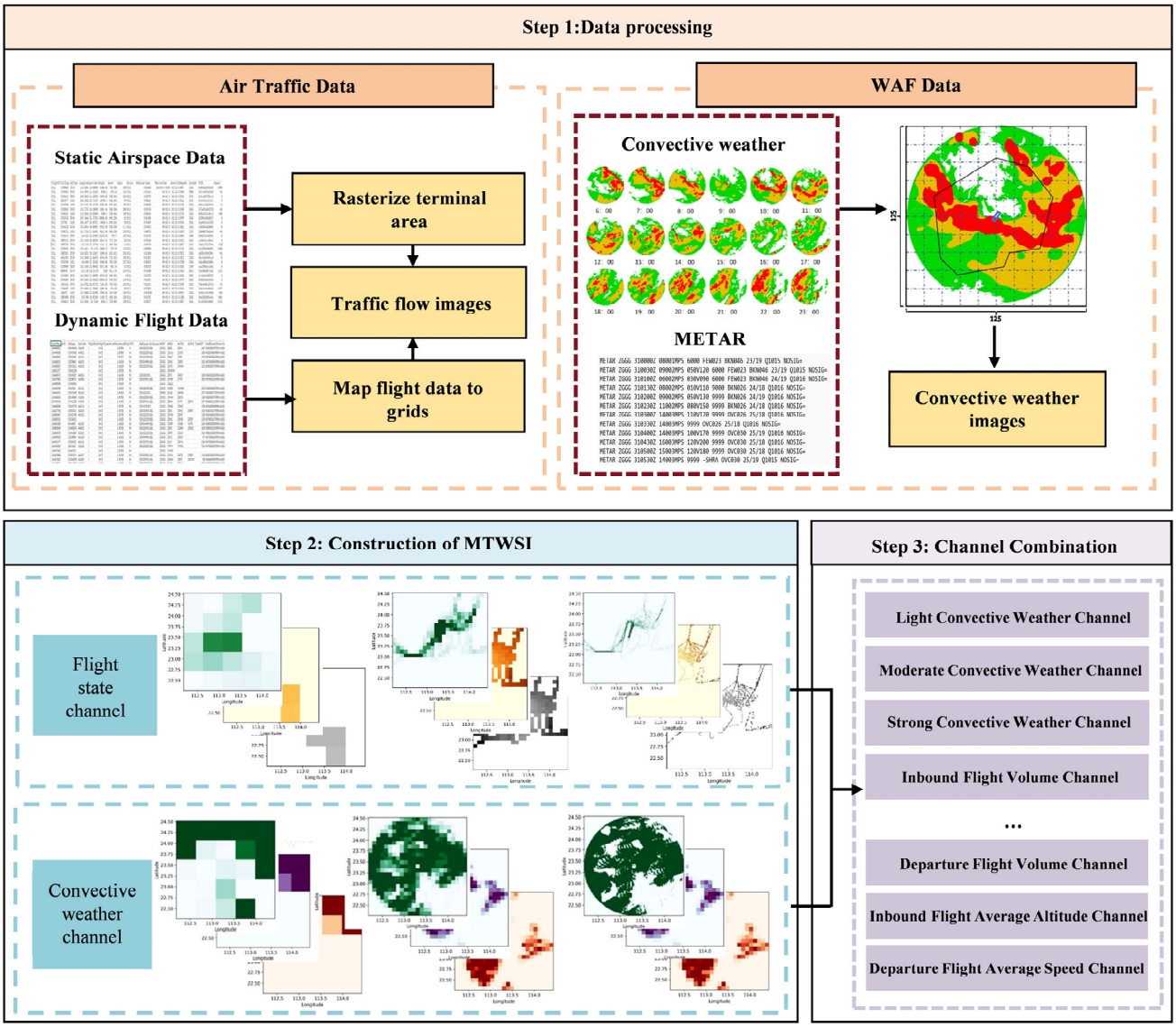


Figure 1. Procedural Framework for Constructing MTWSI.

Each image is represented as a two-dimensional matrix. Therefore, the target terminal area is first rasterized to serve as the foundation for subsequent image construction [26]. To ensure consistency and ease of processing, the outer square boundary of the terminal area is selected as the image extent and divided into 125 × 125 grids (each 2 km wide) (see Figure 2). Since each grid cell contains latitude and longitude information, the aircraft's location within the airspace can be assigned to the corresponding grid, which allows us to populate the grid with traffic operation information, such as traffic volume, average flight altitude, average flight speed, etc. However, because each grid cell can store only one value, multiple 2D matrices are employed to represent diverse traffic and weather information. These various 2D matrices together form a multi-channel image that reflects various aspects of traffic and weather conditions in the same area. By combining these channels, a realistic picture of the air traffic situation is created.

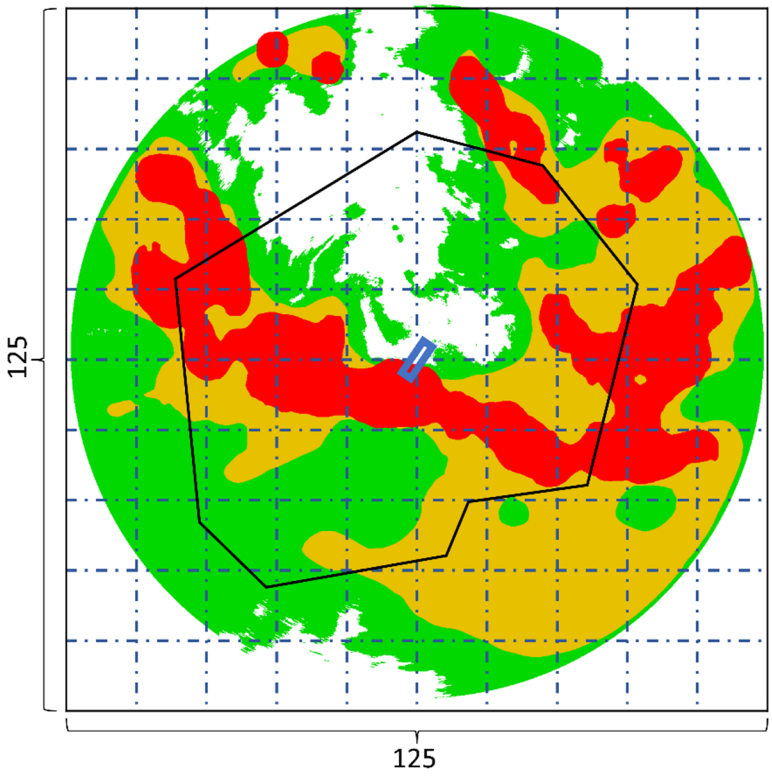


Figure 2. Example of terminal area rasterization.

(2) Visualization of traffic flow features

Since traffic scenarios span a period rather than a single moment, and delay levels are not instantaneous, we aggregate all traffic and convective weather data over the designated period onto a grid map. Specifically, by assigning the flight track data to the corresponding grids of the 2D matrix, the visualization will display the number of flights passing through each grid during a certain period of time, as well as the overall flight status of the flights. Three types of images are based on flight trajectory, altitude, and speed information. These are referred to as the flight volume, average altitude, and average speed channels, as shown in Figure 3. Figure 3a shows the flight volume channel, where a darker grid color indicates that more flights passed through the corresponding airspace during the time period. Figure 3b shows the average altitude channel, where darker colors represent a higher average flight altitude in that grid area. Subfigure (c) shows the average speed channel, where darker colors indicate a lower average flight speed in the corresponding airspace during the time period.

(3) Visualization of convective weather features

In addition, a convective weather channel is constructed to represent weather conditions within the terminal area. The original convective weather WAF data are a two-dimensional matrix consisting of 1250 × 1250 data points. However, its resolution does not match that of the traffic-related channels and therefore needs to be reduced. To address this, following the concept of rasterized WSI, the WAF data are mapped onto a 125 × 125 terminal area grid. The Weather Severity Index (WSI) is then calculated for each grid cell, aligning the data dimensions while preserving both the intensity and spatial distribution of the convective weather. As shown in Figure 4, Figure 4a–c illustrate the distribution of weak, moderate, and strong convective weather within the terminal area, respectively.

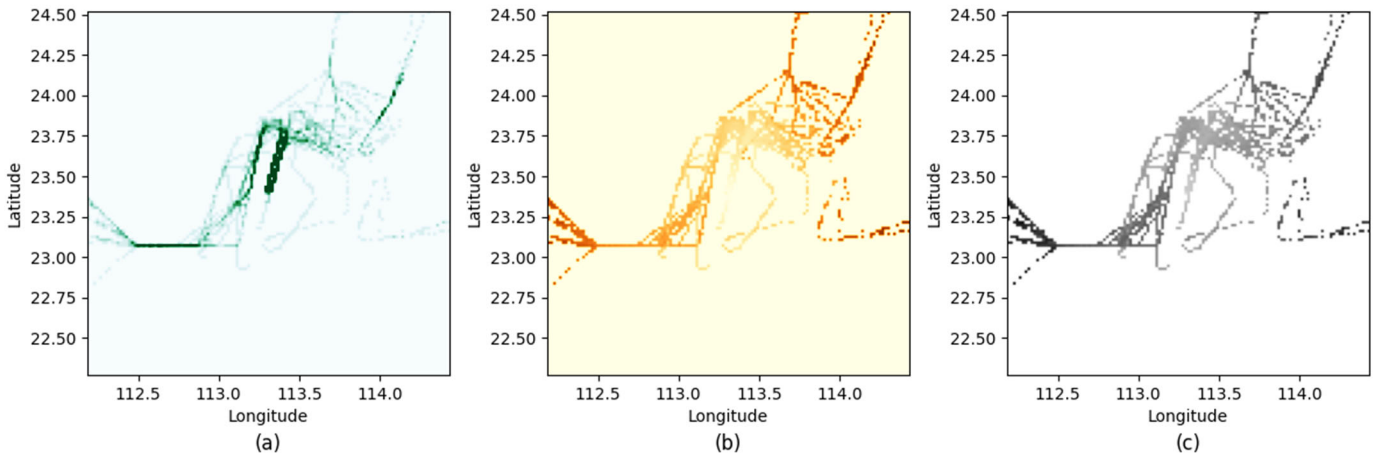


Figure 3. Flight state channel. (**a**) Shows the flight volume channel, (**b**) Shows the average altitude channel, and (**c**) Shows the average speed channel.

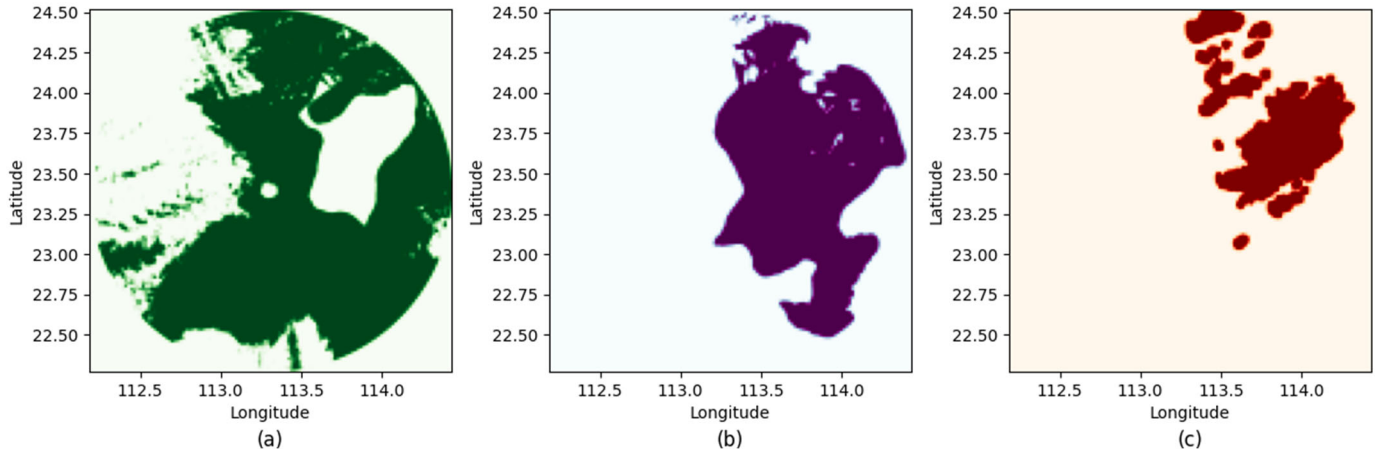


Figure 4. Convective weather channel. (**a**) Shows the weak convective weather within the terminal area, (**b**) Shows the moderate convective weather within the terminal area, (**c**) Shows the strong convective weather within the terminal area.

3.1.3. Construction of Channels

Based on the above channel visualization results, nine image data channels are constructed. These include inbound flight volume channel, inbound flight average flight speed channel, inbound flight average flight altitude channel, departing flight volume channel, departing flight average flight speed channel, departing flight average flight altitude channel, mild convective weather channel, moderate convective weather channel, and strong convective weather channel. In order to eliminate the image features that do not have any significance to the airport delay prediction model, this paper adopts the approximate ablation experiment to screen the channels with various combinations of the channels, and the image resolution is set to 25×25, and the model performance is evaluated by the Accuracy, Precision, Recall, and F1 metrics. Given that the severity of convective weather is a key factor affecting airport delays, all channel combinations must include the three convective weather channels. In total, 145 combinations are tested, and Table 2 lists 8 combinations that achieved better prediction results.

Table 2. Channel Combination.

Combination Name	Channels
Chanel_combi1	Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi2	Inbound Flight Volume, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi3	Departure Flight Volume, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi4	Inbound Flight Volume, Departure Flight Volume, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi5	Inbound Flight Volume, Inbound Flight Average Altitude, Departure Flight Volume, Light Convective Weather, Strong Convective Weather, Moderate Convective Weather
Chanel_combi6	Inbound Flight Volume, Inbound Flight Average Altitude, Inbound Flight Average Speed, Departure Flight Volume, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi7	Inbound Flight Volume, Inbound Flight Average Altitude, Inbound Flight Average Speed, Departure Flight Volume, Departure Flight Average Speed, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather
Chanel_combi8	Inbound Flight Volume, Inbound Flight Average Flight Height, Inbound Flight Average Flight Speed, Departure Flight Volume, Departure Flight Average Flight Speed, Departure Flight Average Flight Height, Light Convective Weather, Moderate Convective Weather, Strong Convective Weather

3.2. CNN-Based Airport Delay Level Prediction Model

Considering image-based methods, we employ a CNN learning strategy to develop an end-to-end method for airport delay prediction, referred to as ADLCNN. Specifically, the raw weather and traffic data are first rasterized to construct MTWSIs. Then, the CNN extracts useful information from these images through nonlinear transformations, completing the feature learning process. Finally, the learned features are used to evaluate the delay level. Figure 5 shows the overall framework of the proposed ADLCNN method.

3.2.1. Classification Method

Airport delay level provides an overall description of various types of delays occurring at an airport within a specific period. It serves as an important indicator for evaluating airport operational efficiency. Using a single feature to describe delay can only reflect one aspect of the situation and fails to capture the full picture of flight delays and their evolving patterns. Therefore, this paper selects two indicators—the average departure delay time and the on-time departure rate—to assess the airport departure delay level from the perspectives of both delay severity and flight reliability.

Airport delay classification can be regarded as an unsupervised learning problem, so in this paper, we will use the fuzzy C-mean (FCM) clustering [27]. The principle of the FCM algorithm is to minimize the criterion function through continuous iteration. After FCM cluster analysis, the flight delay dataset can be divided into different clusters, each representing a level of delay. Therefore, each cluster can be regarded as a delay level, which can be used to classify the degree of delay at the airport. Figure 6 demonstrates the trend of the contour coefficient with the number of clusters, where the number of clusters of 2 corresponds to the highest contour coefficient, indicating that the samples in the clustering results are more cohesive and separated at this time. Figure 7 demonstrates the distribution of different delay features within each cluster when the number of clusters is 2, in which the distribution of average departure delays between two clusters is relatively more separated, while there is a staggered distribution of flight punctuality, but the overall flight punctuality of the samples in Cluster 0 is higher, and the overall flight punctuality of the samples in Cluster 1 is lower.

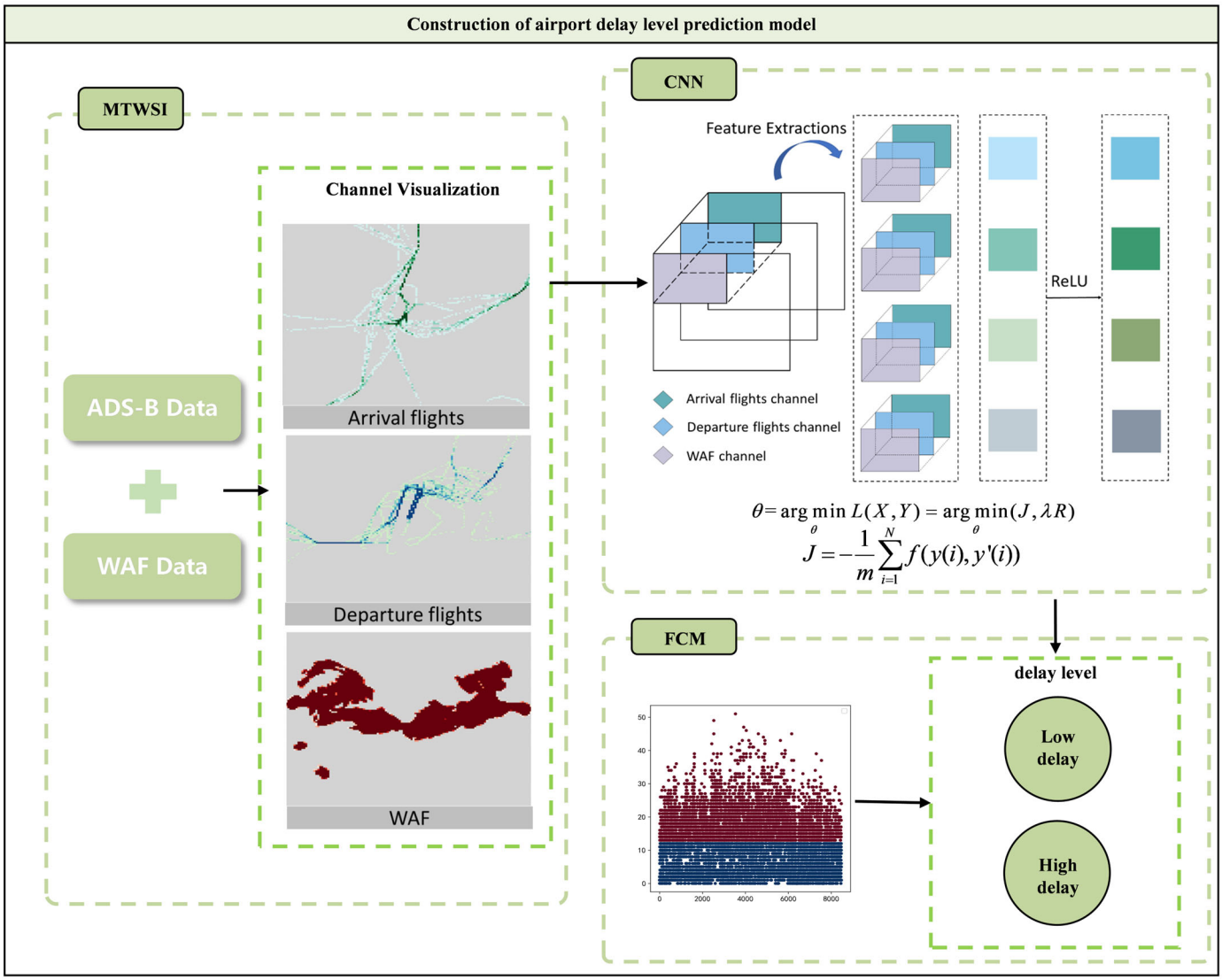


Figure 5. Framework structure of the ADLCNN method.

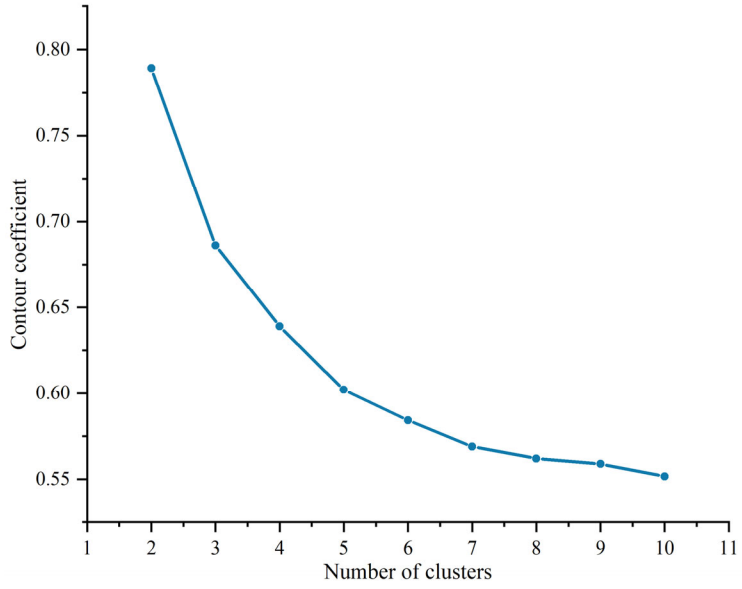


Figure 6. Trend in the number of clusters and profile coefficient.

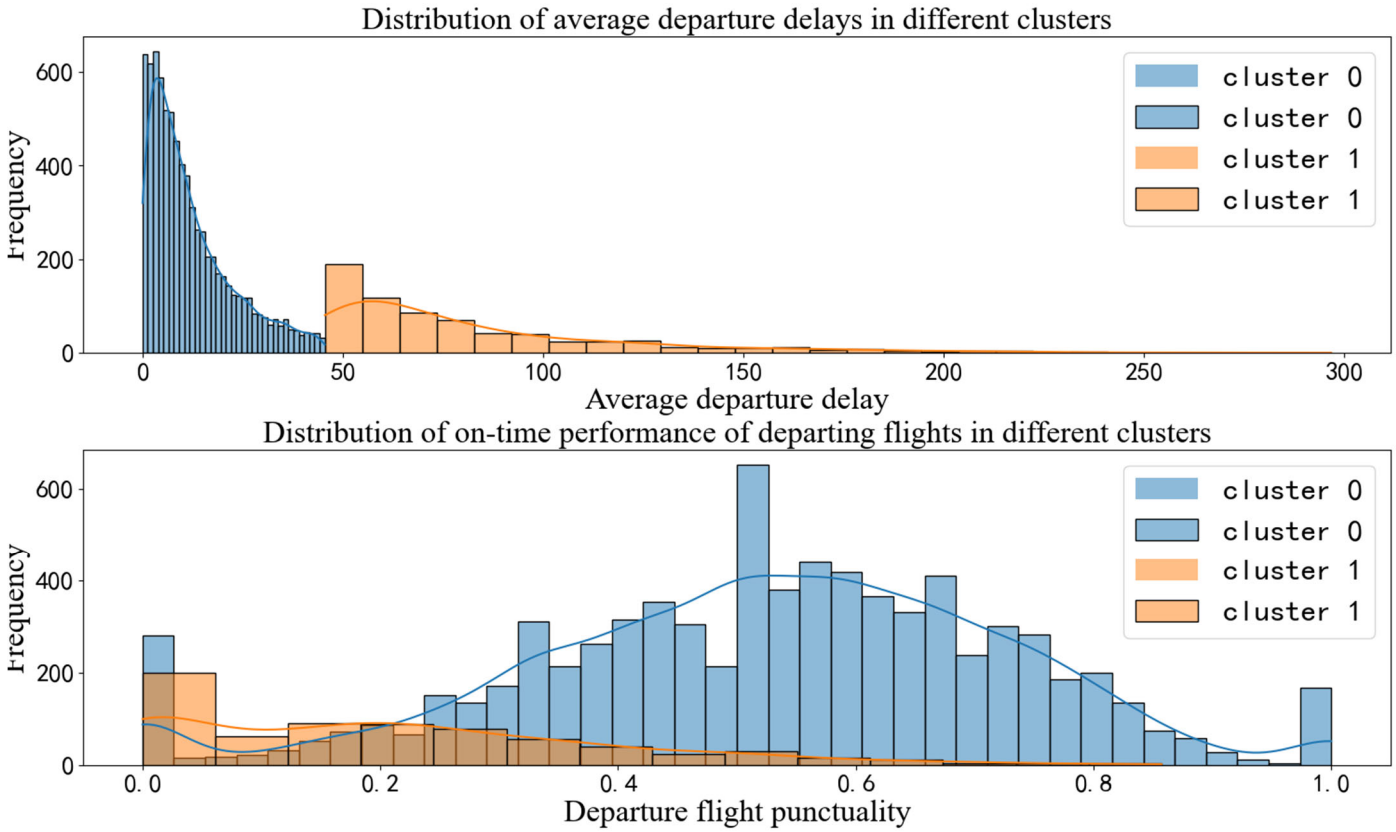


Figure 7. Distribution of delay features in different clusters.

Thus, we divide the airport delay level into two categories. Cluster 0 represents the general delay level, whose average departure delay is less than 50 min, and flight punctuality is higher. Cluster 1 represents the severe delay level, whose average departure delay is higher than 50 min, and flight punctuality is lower. The later experiments on airport delay prediction will be viewed as dichotomous predictions.

The predicted delay levels in this study not only serve as a classification target but also provide practical value for air traffic control by enabling controllers to receive early warnings. For example, if a high delay level is predicted in advance, controllers can prepare diversion resources or adjust flight release priorities accordingly, thereby mitigating congestion and improving decision efficiency.

3.2.2. CNN-Based Feature Extraction and Rank Prediction

This model predicts airport delay levels based on convective weather and air traffic conditions. It utilizes a multi-channel weather-traffic image (MTWSI) generated by combining relevant data. The output corresponds to delay categories created through FCM clustering analysis.

During the construction of feature maps, the entire image is scanned by a processing unit, and the state of this unit is recorded at the corresponding position in the feature map. This process is equivalent to a convolution operation followed by a bias addition and a sigmoid activation function, represented as follows:

$$y^{(d)} = \sigma\left(Wy^{(d-1)} + b\right) \tag{1}$$

where d denotes the depth of the convolutional layer, W is the weight matrix, and b is the bias term. In fully connected neural networks, the weight matrix is dense, meaning that each input is connected to every neuron via distinct weights. However, in convolutional

neural networks (CNNs), due to weight sharing, the weight matrix W is sparse. Its structure can be described as follows:

$$\begin{bmatrix} w & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & w \end{bmatrix} \tag{2}$$

Here, w is a matrix with dimensions matching the receptive field of the unit. The use of sparse weight matrices reduces the number of trainable parameters, thereby enhancing the model's generalization capability. Multiplying the input by the weight matrix W is equivalent to convolving the input with a learnable filter w.

If the input to the $(d-1)$-th layer has a spatial size of N × N, and the receptive field of the units in layer d is m × m, then the resulting feature map will have dimensions (N − m + 1) × (N − m + 1). Specifically, the value at position (i, j) in the feature map is calculated as follows:

$$y_{ij}^{(d)} = \sigma\left(x_{ij}^{(d)} + b\right) \tag{3}$$

$$x_{ij}^{(d)} = \sum_{\alpha=0}^{m-1} \sum_{b=0}^{m-1} w_{ab} y_{(i+a)(j+b)}^{(d-1)} \tag{4}$$

where the bias b is a scalar. By applying Equations (3) and (4) across all input positions (i, j), the feature map for the corresponding plane is generated.

The CNN model contains two convolution layers and three dense layers. The first convolutional layer employs six 3 × 3 kernels, and the second uses sixteen 3 × 3 kernels to extract features. A ReLU activation function follows each dense layer (with 120 and 84 nodes, respectively), and the final layer consists of 2 nodes corresponding to the two classes—delay and no delay—with probabilities computed via a SoftMax function. The probability of each category is calculated using a SoftMax activation function. The model has a learning rate of 0.001 and a batch size of 128. The data distribution is learned from training-set samples by iteratively optimizing the network model parameters. Usually, the optimization direction is determined by the objective function, which consists of an error term (J) and a regularity term (R) [28]:

$$\theta = argminL(X, Y) = argmin(J, \lambda R) \tag{5}$$

$$J = -\frac{1}{m} \sum_{i=1}^{N} f\left(y(i), y'(i)\right) \tag{6}$$

In the equation, X and Y are the inputs and outputs of the model. The loss function and regularization term are denoted by J and R, respectively. θ denotes the parameters of the deep neural network, and λ is the weight of the regularization term. In this paper, the cross-entropy is used as the loss function and is determined by $f(\cdot)$where $y(i)$ and $y\prime\ (i)$ are the real and actual labels of the first i training sample. The use of a shedding layer effectively reduces the occurrence of overfitting, and the Adam optimizer is applied to improve the performance of the gradient descent algorithm. Figure 8 demonstrates the CNN Architecture.

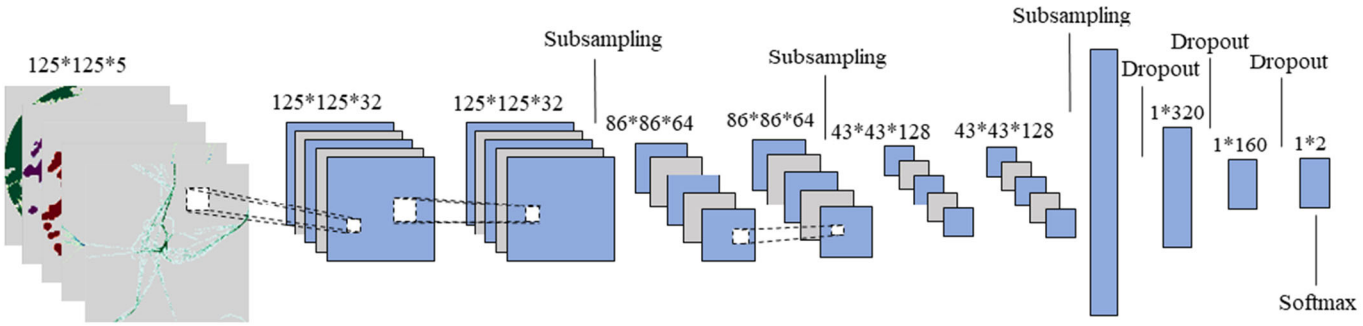


Figure 8. CNN Architecture.

4. Experiment and Results

4.1. Experimental Setup

In this section, we validate the effectiveness of the proposed deep metric learning-based airport delay prediction method using real-world data. The experiments are conducted with one year (2019) of WAF and air traffic data from Guangzhou Baiyun Airport (ZGGG). Each sample represents an MTWSI constructed from one-hour segments of WAF and air traffic data, along with delay level labels derived according to the classification rules described in Section 3.2. In the following experiments, the dataset is divided into training and test sets in a 7:3 ratio. Additionally, two sets of comparison experiments are conducted. The first compares the prediction accuracy of the proposed method with that of a traditional numerical feature processing approach. The second evaluates how different separation rates in multi-channel images affect prediction accuracy.

The experiments were performed on a computer equipped with an AMD EPYC 7642 48-core processor, 24 GB RAM, and an NVIDIA RTX A5000 GPU. The Python programming language, version 3.7.9, was used. Deep learning frameworks TensorFlow 2.4.1 and Keras 2.4.3 were used for model construction and training. Data were processed and analyzed using the NumPy 1.19.5 and pandas 1.2.1 libraries. Machine learning tasks were implemented using the scikit-learn 0.24.1 library. The results were visualized using Matplotlib 3.3.4 and Seaborn 0.11.1 libraries. The experiments were conducted on Ubuntu 20.04.

The predictive model is evaluated using accuracy, precision, recall, F1 score, and Matthews correlation coefficient.

The raw dataset is highly unbalanced and significantly affects the performance of the classifier. To overcome this problem, this study will use sampling techniques to improve the category distribution and avoid overlap. Sampling techniques include under-sampling, oversampling, and hybrid methods. Currently, popular sampling techniques include the following:

(1) Under-sampling (US): Random Under-Sampling (RUS), Edited Nearest Neighbors (ENN), and Tomek Links.

(2) Oversampling (OS) [29,30]: Random Over-Sampling (ROS), Synthetic Minority Over-sampling Technique (SMOTE), and Adaptive Synthetic (ADASYN).

(3) Combination (UOS) [29,30]: SMOTETomek and SMOTEENN

Figure 9 illustrates the original training dataset and the sampled training dataset generated by various sampling techniques to generate the sampled training dataset. The sampled data are used to train the estimation method, and the original test set is used to validate the performance.

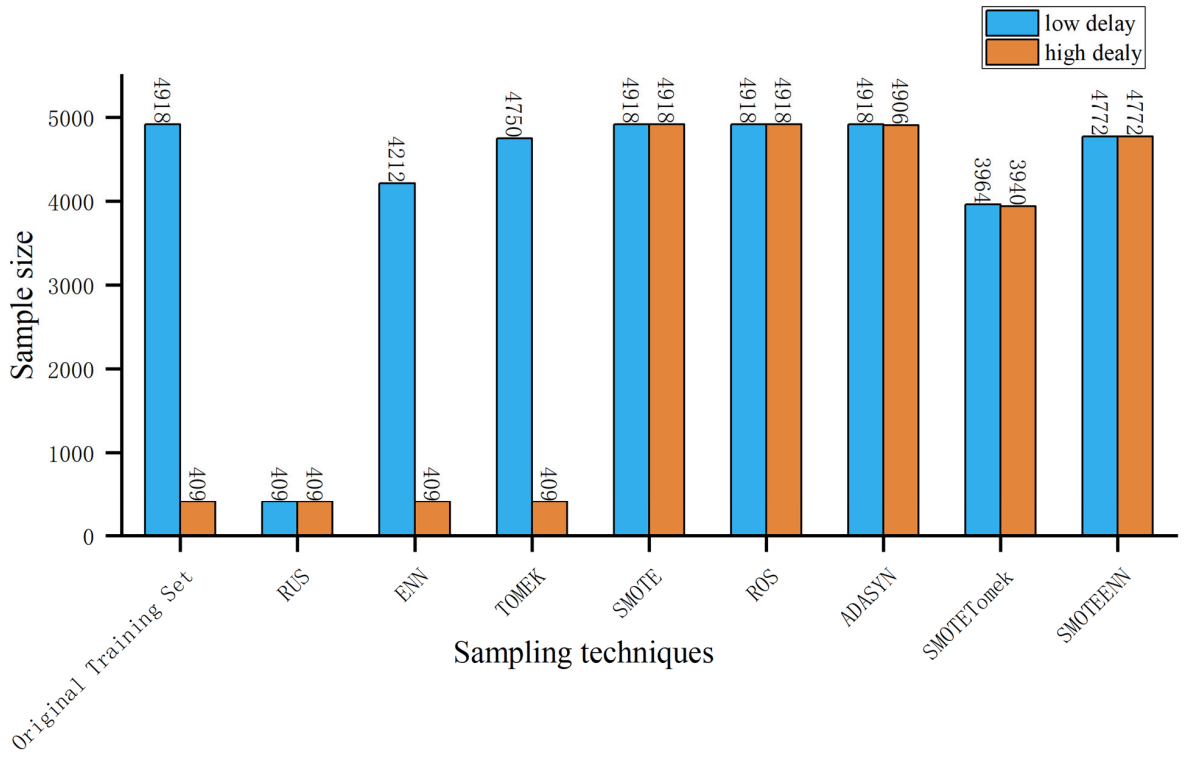


Figure 9. Class distribution of the training set after applying various sampling techniques.

4.2. Results and Discussions

4.2.1. Impact of Channel Selection on Model Performance

Table 3 presents the performance of the ADLCNN model across eight channel combinations for predicting cumulative departure delay levels. Notably, the Chanel_combi7 combination achieves a prediction accuracy of 90.82%, with superior precision, recall, and F1 scores compared to other combinations. Hence, this combination is selected for subsequent predictions. 8 channels: inbound flight volume channel, inbound flight average flight altitude channel, inbound flight average flight speed channel, departing flight volume channel, departing flight average flight speed channel, mild convective weather channel, moderate convective weather channel, and strong convective weather channel.

Table 3. Impact of different channel combinations on cumulative delay level prediction.

Combination Name	Accuracy	Precision	Recall	F1
Chanel_combi1	88.72%	0.86	0.89	0.86
Chanel_combi2	88.85%	0.86	0.89	0.86
Chanel_combi3	90.03%	0.88	0.90	0.89
Chanel_combi4	90.49%	0.89	0.90	0.88
Chanel_combi5	89.84%	0.89	0.90	0.89
Chanel_combi6	90.56%	0.89	0.91	0.89
Chanel_combi7	90.82%	0.90	0.91	0.90
Chanel_combi8	89.70%	0.89	0.90	0.89

4.2.2. Performance Comparison of Delay Prediction Methods

The initial experiment compares the predictive performance of the MTWSI-based CNN model with several machine learning methods that rely on numerical features. These machine learning methods used for comparison include Gaussian Naive Bayes (GNB), k-Nearest Neighbors (KNN), logistic Linear Regression (LLR), Support Vector Machines (SVM), Multi-Layer Perception (MLP), and integrated learning algorithms such as Random Forests (RF) and Adaptive Boost (AdaBoost) [31,32].

Table 4 summarizes the results produced by the prediction methods using various sampling techniques. An in-depth analysis of the table shows that the ADLCNN method proposed in this paper outperforms other traditional machine learning methods overall, regardless of the sampling technique used. In particular, the prediction performance of the ADLCNN is more balanced when using the ROS sampling technique, where the recall is 0.83 for average delays and 0.78 for severe delays, with an overall accuracy of up to 0.831.

Table 4. Prediction results under different sampling methods.

Sampling Methods	Prediction Methods	Accuracy	Label	Classification Report			Confusion Matric	
				Precision	Recall	F1	Low	High
RUS	GNB	0.873	General delay	0.94	0.92	0.93	1942	167
			serious delay	0.24	0.30	0.26	123	52
	KNN	0.673	General delay	0.96	0.68	0.79	1428	681
			serious delay	0.14	0.63	0.23	65	110
	LLR	0.720	General delay	0.97	0.72	0.83	1514	595
			serious delay	0.18	0.75	0.29	44	131
	SVM	0.680	General delay	0.97	0.67	0.80	1421	688
			serious delay	0.16	0.75	0.26	44	131
	MLP	0.699	General delay	0.97	0.70	0.81	1470	639
			serious delay	0.16	0.72	0.27	49	126
	RF	0.735	General delay	0.97	0.73	0.84	1548	561
			serious delay	0.19	0.74	0.30	45	130
	AdaBoost	0.733	General delay	0.98	0.73	0.83	1538	571
			serious delay	0.19	0.78	0.31	39	136
	CNN	0.780	General delay	0.98	0.78	0.87	1643	468
			serious delay	0.23	0.81	0.36	33	142
ROS	GNB	0.871	General delay	0.94	0.92	0.93	1939	170
			serious delay	0.23	0.29	0.26	124	51
	KNN	0.854	General delay	0.94	0.90	0.92	1891	218
			serious delay	0.21	0.34	0.26	116	59
	LLR	0.722	General delay	0.97	0.72	0.83	1522	587
			serious delay	0.18	0.72	0.28	49	126
	SVM	0.684	General delay	0.97	0.68	0.80	1429	680
			serious delay	0.16	0.77	0.27	41	134
	MLP	0.765	General delay	0.97	0.77	0.86	1618	491
			serious delay	0.21	0.74	0.32	46	129
	RF	0.898	General delay	0.94	0.95	0.94	2010	99
			serious delay	0.29	0.23	0.25	135	40
	AdaBoost	0.793	General delay	0.97	0.80	0.88	1685	424
			serious delay	0.23	0.72	0.35	49	126
	CNN	0.831	General delay	0.98	0.83	0.90	1762	349
			serious delay	0.28	0.78	0.41	38	137
SMOTEENN	GNB	0.863	General delay	0.94	0.91	0.92	1914	195
			serious delay	0.23	0.33	0.27	117	58
	KNN	0.786	General delay	0.95	0.81	0.87	1702	407
			serious delay	0.19	0.53	0.28	82	93
	LLR	0.717	General delay	0.97	0.72	0.82	1508	601
			serious delay	0.18	0.74	0.29	45	130
	SVM	0.696	General delay	0.97	0.69	0.81	1456	653
			serious delay	0.17	0.76	0.28	42	133
	MLP	0.806	General delay	0.97	0.82	0.89	1727	382
			serious delay	0.23	0.66	0.34	60	115
	RF	0.820	General delay	0.96	0.84	0.90	1769	340
			serious delay	0.23	0.59	0.33	72	103
	AdaBoost	0.791	General delay	0.97	0.80	0.88	1687	422
			serious delay	0.22	0.69	0.33	55	120
	CNN	0.773	General delay	0.98	0.77	0.86	1621	490
			serious delay	0.23	0.84	0.36	28	147

4.2.3. Performance Comparison of Multi-Channel Images with Different Resolutions

The second experiment compares the performance of ADLCNN models trained on multi-channel images of different resolutions. Here, we have designed three different resolutions, namely 5 × 5, 25 × 25, and 125 × 125.

Table 5 shows the performance results of the ADLCNN method for predicting the cumulative delay level of departures for different resolutions of multi-channel images. Here, the model uses a channel combination of Chanel_combi7 as input. At a 25 × 25 resolution, the model achieves an overall accuracy of 91.61% and a recall of 0.98 for the majority class (A0). However, class imbalance leads to lower accuracy for the minority class (A1), with only 73 out of 172 A1 samples correctly predicted, so the subsequent experiments of comparing the performances of the different methods are carried out.

Table 5. Prediction results of cumulative delay on multichannel images of different resolutions.

Resolution	Accuracy	Label	Classification Report			Confusion Matric	
			Precision	Recall	F1	A0	A1
5 × 5	89.25%	A0	0.91	0.98	0.94	1323	30
		A1	0.56	0.22	0.32	134	38
25 × 25	91.61%	A0	0.93	0.98	0.95	1324	29
		A1	0.72	0.42	0.53	99	73
125 × 125	91.21%	A0	0.92	0.99	0.95	1333	20
		A1	0.74	0.34	0.47	114	58

Table 6 reports the average processing time per step during training and testing at various image resolutions. The computational cost increases dramatically with resolution at 125 × 125, training steps average 238 ms—over 25 times longer than at 25 × 25—and testing steps average 58 ms, nearly 18 times longer. Considering both predictive performance and efficiency, a 25 × 25 resolution is optimal.

Table 6. Model training duration at different resolutions.

	5 × 5	25 × 25	125 × 125
Training time	2 ms/step	9 ms/step	238 ms/step
Testing time	1 ms/step	3 ms/step	58 ms/step

5. Conclusions

This study introduces a multi-channel, image-based method that automatically captures complex, weather-affected traffic features to predict airport delay. The proposed method comprises three main components:

(1) Fuzzy C-Means (FCM) clustering is applied to classify airport departure delay levels.

(2) Weather Advective Flow (WAF) data, combined with incoming and outgoing flight data, are rasterized to construct a multi-channel image that portrays air traffic operation scenarios under convective weather conditions.

(3) Deep learning techniques analyze features in the generated images to predict airport delay levels. Experimental results demonstrate that the proposed method achieves significantly higher prediction accuracy compared to traditional machine learning methods.

The proposed model significantly enhances the accuracy of delay prediction and offers effective support for airport operational decisions. It facilitates proactive resource allocation, schedule adjustments, and reduced passenger waiting times. Furthermore, the model proves highly applicable in real-world scenarios, particularly in capturing the impact of complex weather on air traffic. Its ability to improve operational efficiency

and reduce management costs underscores its strong practical value and potential for broader adoption.

Next steps could explore these key areas:

(1) Designing more complex and effective networks to further enhance prediction performance.

(2) Integrating clustering methods with domain knowledge and professional experience to improve explanatory power in airport delay classification.

(3) Incorporating advanced temporal feature extraction techniques to better capture the time-dependent characteristics inherent in air traffic scenarios.

(4) Considering the method's integration with existing operational protocols such as those of ICAO or local air traffic management systems, the current approach remains at the research stage and has not yet been embedded in formal procedures.

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